NILS JÖRGENSEN

Trustworthy Cyber-Physical Systems Researcher

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EDUCATION

Ph. D. in Mechatronics and Embedded Control Systems
KTH Royal Institute of Technology

Autumn 2020 – Spring 2026 | Stockholm, Sweden

Trustworthy Cyber-Physical Systems • Fault-Tolerant Design • Formal Methods and Software Verification • Cyber-Physical Security • Networked Control Systems • Hybrid Systems • Reinforcement Learning • Convex Optimization • Research Methodology • Scientific Writing and Communication

M. Sc. in Engineering Design – Mechatronics Track
KTH Royal Institute of Technology

Autumn 2017 – Spring 2019 | Stockholm, Sweden

Internetworking • Internet Security and Privacy • Embedded and Distributed Control Systems • Nonlinear Control • Optimal and Model-Predictive Control • Low-Level Programming • Robust Electronics • Test-Driven Development • Systems Engineering

B. Sc. in Mechanical Engineering – International Program
RWTH Aachen University / KTH Royal Institute of Technology

Autumn 2014 – Summer 2017 | Aachen, Germany

Automatic Control • Object-Oriented Programming • Algorithms and Optimisation • Statistics and Probability Theory • Numerical Analysis • Machine Components • Electrical Engineering

EXPERIENCE

Teacher, Student Supervisor, and Research Engineer
KTH | Department of Machine Design

Autumn 2019 – Spring 2026 | Stockholm, Sweden

Years of teaching, course development, and research, covering computer architectures, assembly and bare-metal programming, real-time operating systems, distributed systems, modeling and control of dynamical systems, and everything in between. Supervised students in degree projects as well as industrial prototyping projects within the Mechatronics capstone course.

Summer Internship
Ericsson Research | Device Technologies

Summers 2018 and 2019 | Stockholm, Sweden

Worked two summers developing hardware and software-based platforms for various applications within Augmented Reality.

LANGUAGES

Swedish	●●●●●
English	●●●●●
German	●●●●○

TECHNICAL SKILLS

Trustworthy & secure cyber-physical systems
Safety, reliability, fault tolerance and cyber-security of embedded and networked control systems.

Embedded computing systems
Computer architectures, assembly, bare-metal and RTOS, hardware instrumentation and debugging.

Formal methods & systems theory
Model checking, satisfiability solving, hybrid-systems analysis. Nonlinear and optimal control.

Networked & distributed systems
Wireless communication, 5G/6G, edge computing and physical AI.

PUBLICATIONS

Trustworthy edge computing for cyber-physical systems
10.1145/3539662

Review of communication-aware robot planning algorithms for 5G
10.1109/ETFA52439.2022.9921449

Joint 5G-aware task planner cuts spectrum use by 50% using PDDL
10.1109/ETFA54631.2023.10275420

Dominant models systematically over-predict 5G throughput
10.48550/arXiv.2603.08865

REFERENCES

Available upon request.